

Exploring causal modeling to enhance diabetes prediction and management

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Abstract—Gestational Diabetes Mellitus is a common pregnancy complication, affecting up to 10% of pregnancies worldwide. This study applies causal modeling to PIMA dataset to support diabetes prediction and management. Results show that healthy lifestyle interventions reduce glucose level, and counterfactual analysis suggest that modifying key-factors may prevent disease onset. Findings highlight the potential of causal inference for risk identification and prevention.

Index Terms—Diabetes, Causal Discovery, Intervention, Counterfactual, Pima Indians Diabetes Database

I. INTRODUCTION

Gestational diabetes is pregnancy-induced glucose intolerance due to placental hormones impairing insulin function [1]. To understand diabetes and its risk factors, we propose *structural causal models* (SCM) [2] learned from data. Our methods involves two step: *causal learning* from observational data, avoiding randomized control trials (RCT) limitations; and *causal inference* step, through interventional and counterfactual queries for predictive, and retrospective analysis. The above objectives are formulated as research questions:

- **RQ1**: Which features have a causal effect on diabetes?
- **RQ2**: What are the effect of *interventions* on glucose?
- **RQ3**: What are the *counterfactual* diabetes outcomes?

II. RELATED WORK

The PIMA dataset is widely adopted for its high-risk demographic and inclusion of key physiological and lifestyle factors, supporting feature analysis and the evaluation of diabetes prediction models. Mousa [3] compares three machine learning algorithms on this dataset. Olisah et al. [4] apply Spearman correlation and polynomial regression for missing data handling and feature selection, achieving 97.93% accuracy with a DNN. Liang et al [5] employ Bayesian Networks to evaluate diabetes risk, thus probabilistic dependencies and supporting causal reasoning. In contrast, we aim at learning the causal structure and answering causal queries.

III. METHODS

A. Dataset and Preprocessing

The **Pima Indians Diabetes** dataset [6], contains 768 records of women aged 21-81, with 9 physiological and lifestyle features. Despite no missing values, some field, like *BMI* (Body Mass Index), *Glucose*, *Blood Pressure*, *Insulin*, and *Skin Thickness*, include biologically implausible zeros. We

removed rows with at least one zero in these fields — our only preprocessing step. Without input from domain experts, we avoided outlier handling to minimize bias and preserve informative data.

B. Causal Analysis

Causal models identify and quantify relationships between variables. Structural Causal Models (SCMs) represent these relations as Directed Acyclic Graph (DAG), where each variable depends on its direct causes and a noise component.

Causal Discovery. Causal models can be built using domain expert knowledge, RCTs, or causal discovery (CD) algorithms on observational data [7]. CD infers causal structures from statistical dependencies, and often allows integration of prior knowledge. When domain expertise is lacking, Large Language Models can validate CD results improving robustness and interpretability of causal analyses [8], [9].

Causal Inference. Once the causal model is available, causal inference (CI) estimates the effect of changing a variable on an outcome of interest. Unlike the ML, which relies on correlation identifies ($P(X|W)$), CI focuses on causality. CI aims to answers causal question: *What output X do we expect if we actively set W to w (intervention) and What would have happened to X if we had set W to w ? (counterfactual).* CI supports reasoning about hypothetical scenarios and designing effective interventions [2].

IV. RESULTS

We first report the results of applying CD to the PIMA dataset to answer **RQ1** (Section IV-A). To answer **RQ2** and **RQ3**, we randomly selected four individuals (I), one per overweight class, and defined four age-based subgroups (S). Intervention and counterfactual effects are estimated using DoWhy [10], with 5,000 repetitions per individual and subgroup to account for the probabilistic nature of causal models.

A. RQ1: Causal Discovery

To derive the causal model, we employed the DirectLiNGAM algorithm [11], which captures pairwise relationships via structural equations. However, it misidentified the direction of the *Outcome* variable, suggesting it was influenced only by *Age* and causally affected all other features. We leveraged *ChatGpt-4*, as a domain expert to refine pairwise direction in the causal structure. The final model, shown in

Fig. 1, includes these revisions: all edges were reversed toward Outcome, the arc $Age \rightarrow Pregnancies$ was removed (age affects conception, but not directly the number of pregnancies), and the arc $Pregnancies \rightarrow Outcome$, consistent with the biological mechanisms previously described. This causal graph is aligned with the data, with estimated effects passed key refutation tests with statistical significance ($\alpha = .05$).¹

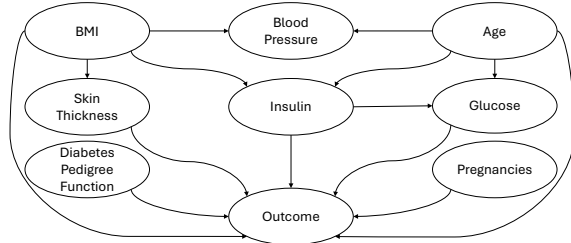


Fig. 1. Causal graph validate with ChatGpt-4

B. RQ2: Intervention

To simulate a healthy lifestyle, we fix BMI at 22 and evaluate its effect on **Glucose**. This choice reflects diabetes management practices, where glucose control is key. Table I shows the effect of the intervention, in terms of variation of **glucose post-intervention** (PG) with respect to the **initial glucose** (IG) values. Most median differences are positive, indicating a general reduction in glucose levels. However, S1 and S3 show negative average differences. For S1, the largest group, the change is minimal, with a confidence interval (CI) nearly including zero. In S3, the small sample size and other factors may explain the observed glucose trends.

Id	Characteristics	IG-PG	95% CI
I1	BMI = 25.2, healthy	4.31	-54.81, 39.78
I2	BMI = 32.9, diabetic	12.10	-47.60, 47.18
I3	BMI = 35.9, diabetic	23.39	-35.67, 58.47
I4	BMI = 40.6, healthy	31.69	-26.23, 67.85
S1	Age = [20-29], d = 48, h = 190	-3.30	-6.49, -0.04
S2	Age = [30-39], d = 35, h = 43	5.69	0.00, 11.17
S3	Age = [40-49], d = 26, h = 20	-8.51	-15.79, -1.76
S4	Age ≥ 50, d = 26, h = 20	19.89	10.96, 28.07

TABLE I
INTERVENTION BMI=22 ON INDIVIDUALS & SUBGROUPS ON 5000
REPET. D: DIABETICS, H: HEALTHY

C. RQ3: Counterfactual

For counterfactuals analysis, we used diabetic individuals I2 and I3, and diabetic patients from subgroups S1–S4. In both cases, we investigated how **Outcomes** would change altering the values of **BMI** or **Pregnancies**. Table II reports the number of cases “#Healthy” (median over repetitions) that would switch from diabetic to healthy in these retrospective scenarios. Both individuals remained diabetic, suggesting other co-causes. Instead, each subgroup included patients who would have become healthy under the counterfactual conditions².

¹Tests: Placebo Treatment; Dummy Outcome; Random Common Cause [12]

²Repetitions here gave consistently the same result (CI equals the median).

Id	#Diabetic	Query	#Healthy	95% CI
S1	48	BMI = 22	8	8, 8
S1	48	Pregnancies = 2	1	1, 1
S2	35	BMI = 22	12	12, 12
S2	35	Pregnancies = 2	3	3, 3
S3	26	BMI = 22	4	4, 4
S3	26	Pregnancies = 2	1	1, 1
S4	21	BMI = 22	6	6, 6
S4	21	Pregnancies = 2	5	5, 5

TABLE II
COUNTERFACTUAL ON SUBGROUPS ON 5000 REPET.

V. CONCLUSION

This study applied causal discovery to the PIMA diabetes dataset to identify factors influencing diabetes onset. Findings show that causal models can reveal risk factors that may help prevent or delay the disease. Intervention analysis on individuals and subgroups, aligned with existing literature, indicating potential for evaluating clinical guidelines and deepening understanding of diabetes. Counterfactual analysis revealed how BMI changes affect diabetes outcomes, including average shifts from diabetic to healthy status. These results underscore the potential of causal inference in guiding prevention strategies. Future work should extend analyses and address dataset limitations with broader, more detailed data.

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